

The University of the State of New York
REGENTS HIGH SCHOOL EXAMINATION

GEOMETRY (Common Core)

Tuesday, June 2, 2015 — 1:15 to 4:15 p.m., only

Student Name: _____

School Name: _____

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

Print your name and the name of your school on the lines above.

A separate answer sheet for Part I has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 36 questions. You must answer all questions in this examination. Record your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in Parts II, III, and IV directly in this booklet. All work should be written in pen, except for graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale.

The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will *not* be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

Notice...

A graphing calculator, a straightedge (ruler), and a compass must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

High School Math Reference Sheet

1 inch = 2.54 centimeters	1 kilometer = 0.62 mile	1 cup = 8 fluid ounces
1 meter = 39.37 inches	1 pound = 16 ounces	1 pint = 2 cups
1 mile = 5280 feet	1 pound = 0.454 kilogram	1 quart = 2 pints
1 mile = 1760 yards	1 kilogram = 2.2 pounds	1 gallon = 4 quarts
1 mile = 1.609 kilometers	1 ton = 2000 pounds	1 gallon = 3.785 liters
		1 liter = 0.264 gallon
		1 liter = 1000 cubic centimeters

Triangle	$A = \frac{1}{2}bh$
Parallelogram	$A = bh$
Circle	$A = \pi r^2$
Circle	$C = \pi d$ or $C = 2\pi r$
General Prisms	$V = Bh$
Cylinder	$V = \pi r^2 h$
Sphere	$V = \frac{4}{3}\pi r^3$
Cone	$V = \frac{1}{3}\pi r^2 h$
Pyramid	$V = \frac{1}{3}Bh$

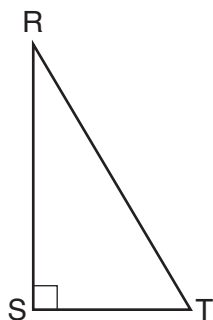
Pythagorean Theorem	$a^2 + b^2 = c^2$
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Arithmetic Sequence	$a_n = a_1 + (n - 1)d$
Geometric Sequence	$a_n = a_1 r^{n-1}$
Geometric Series	$S_n = \frac{a_1 - a_1 r^n}{1 - r}$ where $r \neq 1$
Radians	1 radian = $\frac{180}{\pi}$ degrees
Degrees	1 degree = $\frac{\pi}{180}$ radians
Exponential Growth/Decay	$A = A_0 e^{k(t - t_0)} + B_0$

Part I

Answer all 24 questions in this part. Each correct answer will receive 2 credits. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. [48]

Use this space for
computations.

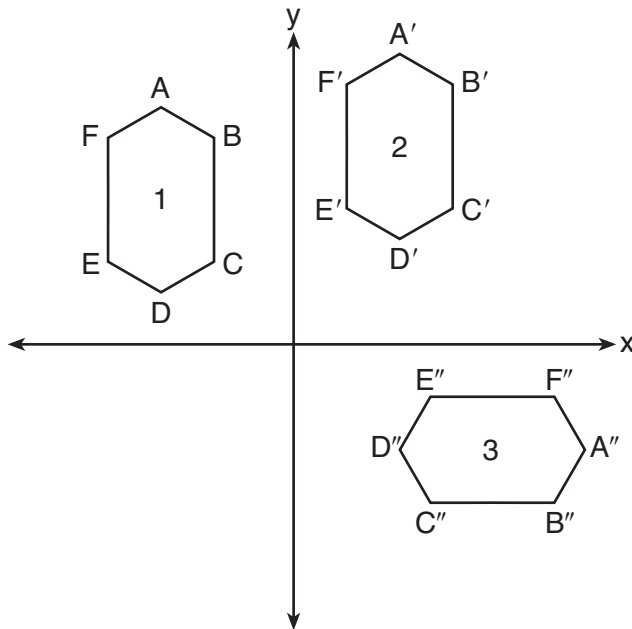
- 1 Which object is formed when right triangle RST shown below is rotated around leg $\overline{RS}$?



- (1) a pyramid with a square base (3) a right triangle
(2) an isosceles triangle (4) a cone
- 2 The vertices of $\triangle JKL$ have coordinates $J(5,1)$, $K(-2,-3)$, and $L(-4,1)$. Under which transformation is the image $\triangle J'K'L'$ *not* congruent to $\triangle JKL$?
- (1) a translation of two units to the right and two units down
(2) a counterclockwise rotation of 180 degrees around the origin
(3) a reflection over the x -axis
(4) a dilation with a scale factor of 2 and centered at the origin
- 3 The center of circle Q has coordinates $(3,-2)$. If circle Q passes through $R(7,1)$, what is the length of its diameter?
- (1) 50 (3) 10
(2) 25 (4) 5

**Use this space for
computations.**

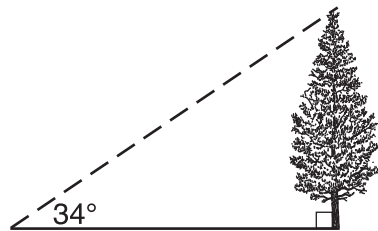
4 In the diagram below, congruent figures 1, 2, and 3 are drawn.



Which sequence of transformations maps figure 1 onto figure 2 and then figure 2 onto figure 3?

- (1) a reflection followed by a translation
- (2) a rotation followed by a translation
- (3) a translation followed by a reflection
- (4) a translation followed by a rotation

5 As shown in the diagram below, the angle of elevation from a point on the ground to the top of the tree is 34° .

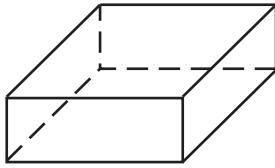


If the point is 20 feet from the base of the tree, what is the height of the tree, to the *nearest tenth of a foot*?

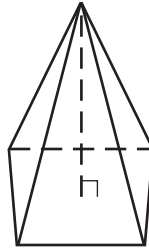
- (1) 29.7
- (2) 16.6
- (3) 13.5
- (4) 11.2

**Use this space for
computations.**

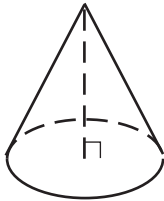
6 Which figure can have the same cross section as a sphere?



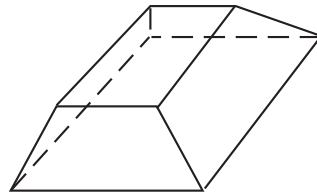
(1)



(3)



(2)



(4)

7 A shipping container is in the shape of a right rectangular prism with a length of 12 feet, a width of 8.5 feet, and a height of 4 feet. The container is completely filled with contents that weigh, on average, 0.25 pound per cubic foot. What is the weight, in pounds, of the contents in the container?

(1) 1,632

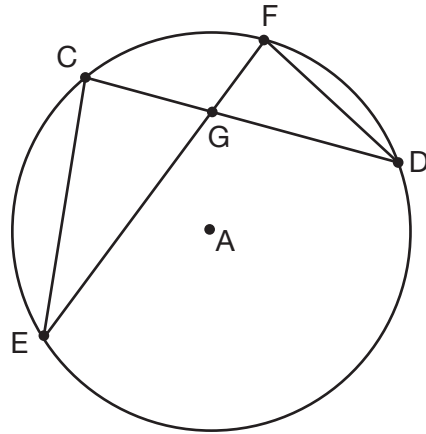
(3) 102

(2) 408

(4) 92

Use this space for
computations.

- 8 In the diagram of circle A shown below, chords $\overline{CD}$ and $\overline{EF}$ intersect at G , and chords $\overline{CE}$ and $\overline{FD}$ are drawn.



Which statement is *not* always true?

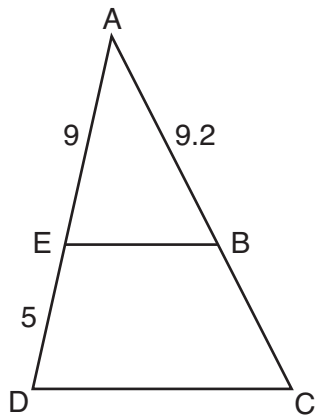
- (1) $\overline{CG} \cong \overline{FG}$ (3) $\frac{CE}{EG} = \frac{FD}{DG}$
(2) $\angle CEG \cong \angle FDG$ (4) $\triangle CEG \sim \triangle FDG$
- 9 Which equation represents a line that is perpendicular to the line represented by $2x - y = 7$?
- (1) $y = -\frac{1}{2}x + 6$ (3) $y = -2x + 6$
(2) $y = \frac{1}{2}x + 6$ (4) $y = 2x + 6$

**Use this space for
computations.**

10 Which regular polygon has a minimum rotation of 45° to carry the polygon onto itself?

- (1) octagon
- (2) decagon
- (3) hexagon
- (4) pentagon

11 In the diagram of $\triangle ADC$ below, $\overline{EB} \parallel \overline{DC}$, $AE = 9$, $ED = 5$, and $AB = 9.2$.

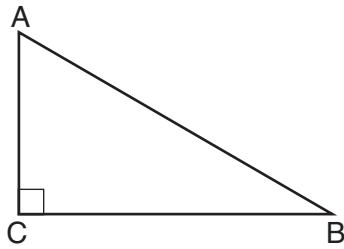


What is the length of $\overline{AC}$, to the *nearest tenth*?

- (1) 5.1
- (2) 5.2
- (3) 14.3
- (4) 14.4

Use this space for
computations.

- 12 In scalene triangle ABC shown in the diagram below, $m\angle C = 90^\circ$.

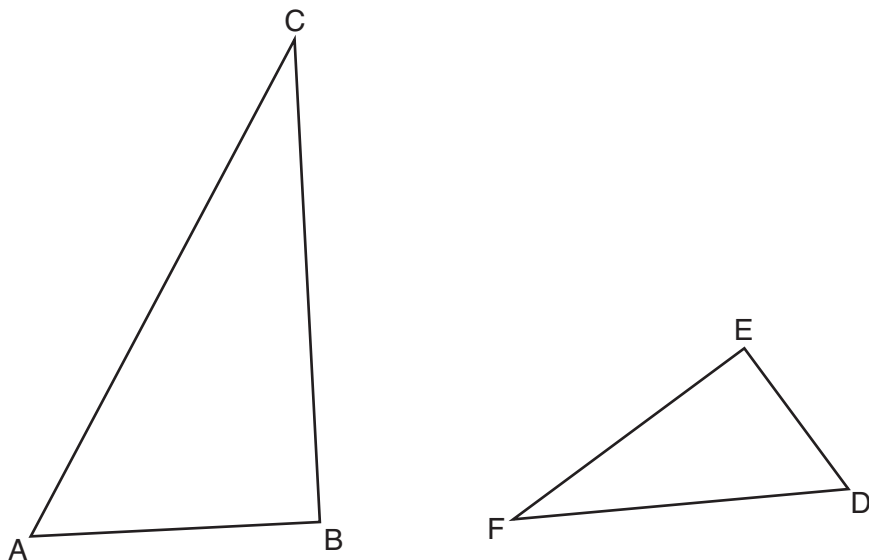


Which equation is always true?

- (1) $\sin A = \sin B$ (3) $\cos A = \sin C$
(2) $\cos A = \cos B$ (4) $\sin A = \cos B$
- 13 Quadrilateral $ABCD$ has diagonals $\overline{AC}$ and $\overline{BD}$. Which information is *not* sufficient to prove $ABCD$ is a parallelogram?
- (1) $\overline{AC}$ and $\overline{BD}$ bisect each other.
(2) $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{AD}$
(3) $\overline{AB} \cong \overline{CD}$ and $\overline{AB} \parallel \overline{CD}$
(4) $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \parallel \overline{AD}$
- 14 The equation of a circle is $x^2 + y^2 + 6y = 7$. What are the coordinates of the center and the length of the radius of the circle?
- (1) center $(0,3)$ and radius 4
(2) center $(0,-3)$ and radius 4
(3) center $(0,3)$ and radius 16
(4) center $(0,-3)$ and radius 16

15 Triangles ABC and DEF are drawn below.

**Use this space for
computations.**



If $AB = 9$, $BC = 15$, $DE = 6$, $EF = 10$, and $\angle B \cong \angle E$, which statement is true?

(1) $\angle CAB \cong \angle DEF$ (3) $\triangle ABC \sim \triangle DEF$

(2) $\frac{AB}{CB} = \frac{FE}{DE}$ (4) $\frac{AB}{DE} = \frac{FE}{CB}$

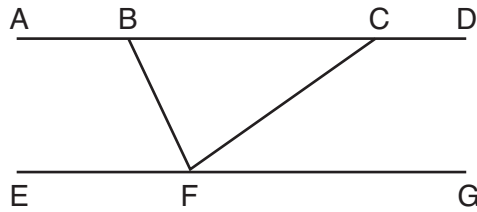
16 If $\triangle ABC$ is dilated by a scale factor of 3, which statement is true of the image $\triangle A'B'C'$?

(1) $3A'B' = AB$ (3) $m\angle A' = 3(m\angle A)$

(2) $B'C' = 3BC$ (4) $3(m\angle C') = m\angle C$

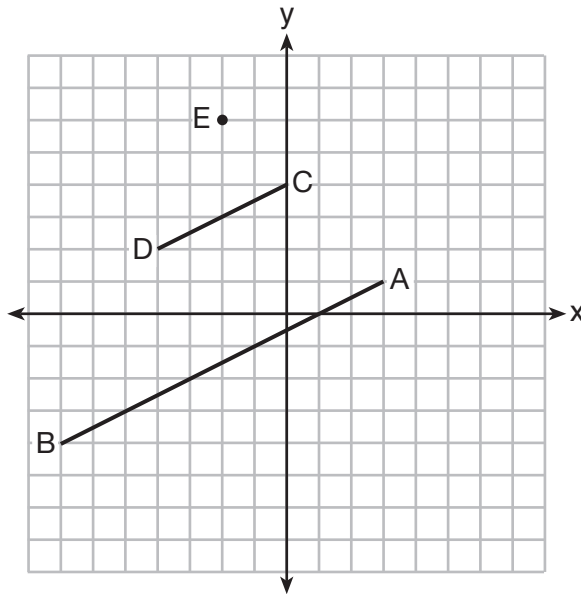
Use this space for
computations.

- 17 Steve drew line segments $ABCD$, EFG , BF , and CF as shown in the diagram below. Scalene $\triangle BFC$ is formed.



Which statement will allow Steve to prove $\overline{ABCD} \parallel \overline{EFG}$?

- (1) $\angle CFG \cong \angle FCB$ (3) $\angle EFB \cong \angle CFB$
 (2) $\angle ABF \cong \angle BFC$ (4) $\angle CBF \cong \angle GFC$
- 18 In the diagram below, $\overline{CD}$ is the image of $\overline{AB}$ after a dilation of scale factor k with center E .



Which ratio is equal to the scale factor k of the dilation?

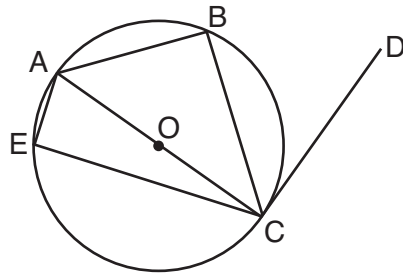
- (1) $\frac{EC}{EA}$ (3) $\frac{EA}{BA}$
 (2) $\frac{BA}{EA}$ (4) $\frac{EA}{EC}$

Use this space for
computations.

- 19** A gallon of paint will cover approximately 450 square feet. An artist wants to paint all the outside surfaces of a cube measuring 12 feet on each edge. What is the *least* number of gallons of paint he must buy to paint the cube?

(1) 1 (3) 3
(2) 2 (4) 4

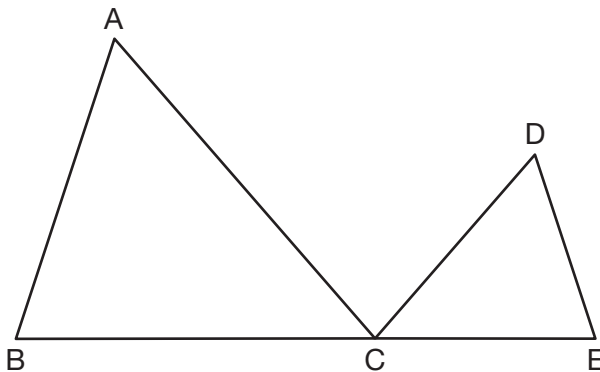
- 20** In circle O shown below, diameter $\overline{AC}$ is perpendicular to $\overline{CD}$ at point C , and chords $\overline{AB}$, $\overline{BC}$, $\overline{AE}$, and $\overline{CE}$ are drawn.



Which statement is *not* always true?

(1) $\angle ACB \cong \angle BCD$ (3) $\angle BAC \cong \angle DCB$
(2) $\angle ABC \cong \angle ACD$ (4) $\angle CBA \cong \angle AEC$

- 21** In the diagram below, $\triangle ABC \sim \triangle DEC$.



If $AC = 12$, $DC = 7$, $DE = 5$, and the perimeter of $\triangle ABC$ is 30, what is the perimeter of $\triangle DEC$?

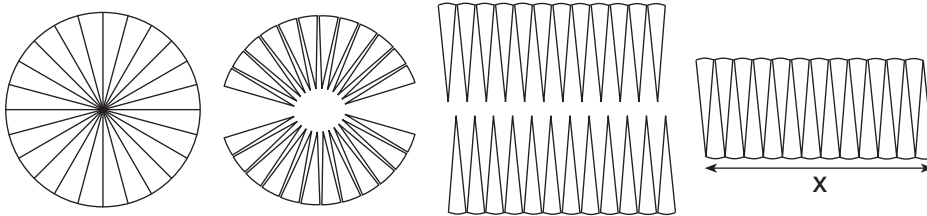
(1) 12.5 (3) 14.8
(2) 14.0 (4) 17.5

Use this space for
computations.

- 22 The line $3y = -2x + 8$ is transformed by a dilation centered at the origin. Which linear equation could be its image?

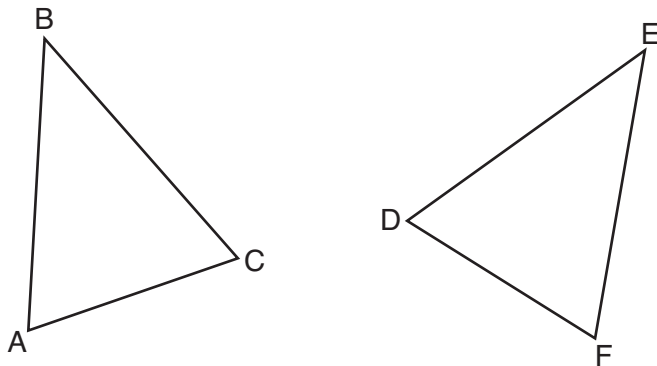
- (1) $2x + 3y = 5$ (3) $3x + 2y = 5$
(2) $2x - 3y = 5$ (4) $3x - 2y = 5$

- 23 A circle with a radius of 5 was divided into 24 congruent sectors. The sectors were then rearranged, as shown in the diagram below.



To the *nearest integer*, the value of x is

- (1) 31 (3) 12
(2) 16 (4) 10
- 24 Which statement is sufficient evidence that $\triangle DEF$ is congruent to $\triangle ABC$?

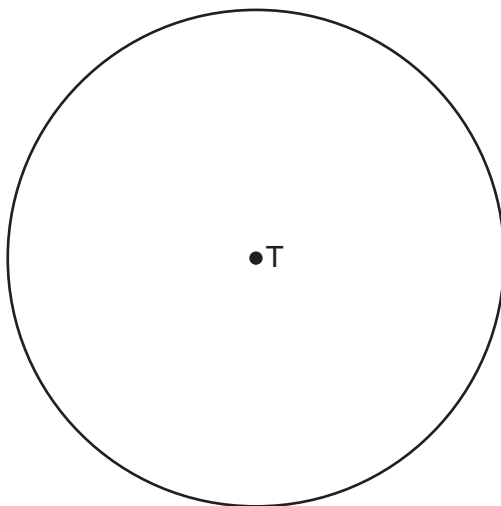


- (1) $AB = DE$ and $BC = EF$
(2) $\angle D \cong \angle A$, $\angle B \cong \angle E$, $\angle C \cong \angle F$
(3) There is a sequence of rigid motions that maps $\overline{AB}$ onto $\overline{DE}$, $\overline{BC}$ onto $\overline{EF}$, and $\overline{AC}$ onto $\overline{DF}$.
(4) There is a sequence of rigid motions that maps point A onto point D , $\overline{AB}$ onto $\overline{DE}$, and $\angle B$ onto $\angle E$.

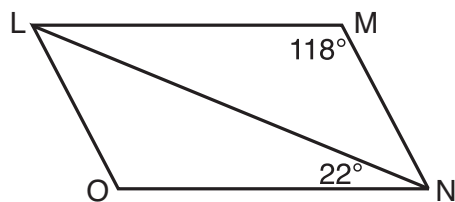
Part II

Answer all 7 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [14]

- 25** Use a compass and straightedge to construct an inscribed square in circle T shown below.
[Leave all construction marks.]

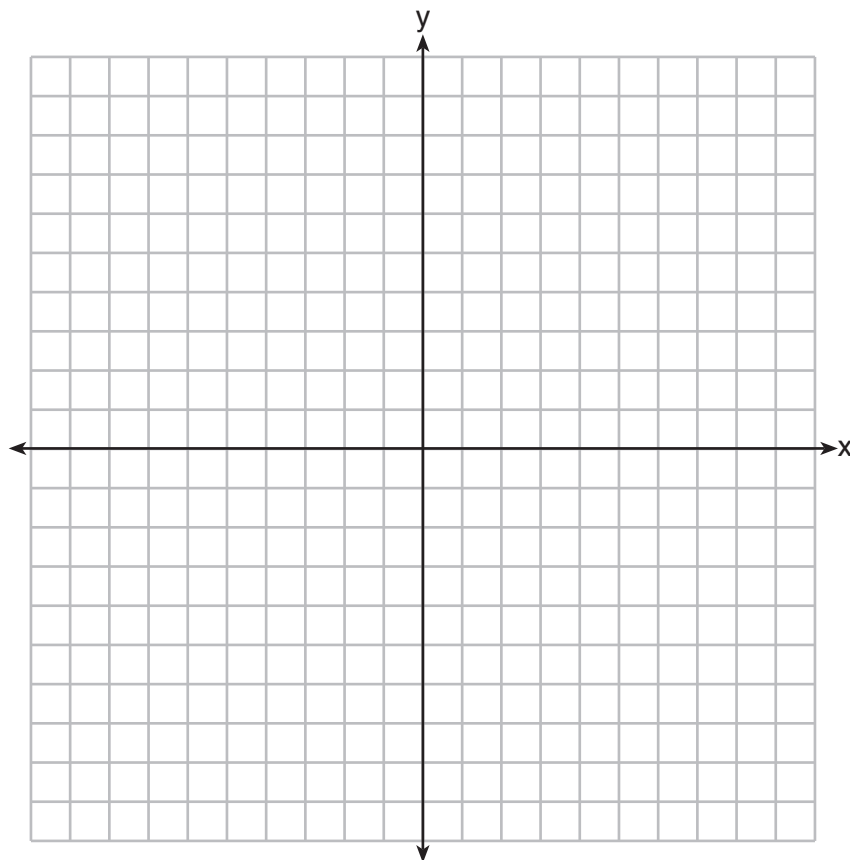


- 26** The diagram below shows parallelogram $LMNO$ with diagonal $\overline{LN}$, $m\angle M = 118^\circ$, and $m\angle LNO = 22^\circ$.

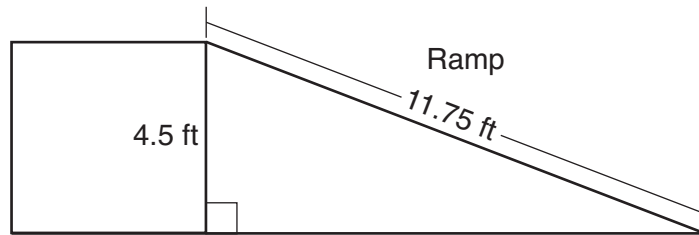


Explain why $m\angle NLO$ is 40 degrees.

- 27** The coordinates of the endpoints of $\overline{AB}$ are $A(-6, -5)$ and $B(4, 0)$. Point P is on $\overline{AB}$. Determine and state the coordinates of point P , such that $AP:PB$ is 2:3.
[The use of the set of axes below is optional.]

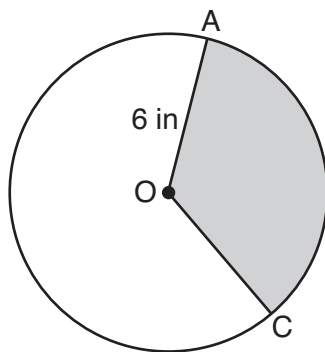


- 28** The diagram below shows a ramp connecting the ground to a loading platform 4.5 feet above the ground. The ramp measures 11.75 feet from the ground to the top of the loading platform.



Determine and state, to the *nearest degree*, the angle of elevation formed by the ramp and the ground.

- 29** In the diagram below of circle O , the area of the shaded sector AOC is $12\pi \text{ in}^2$ and the length of $\overline{OA}$ is 6 inches. Determine and state $m\angle AOC$.



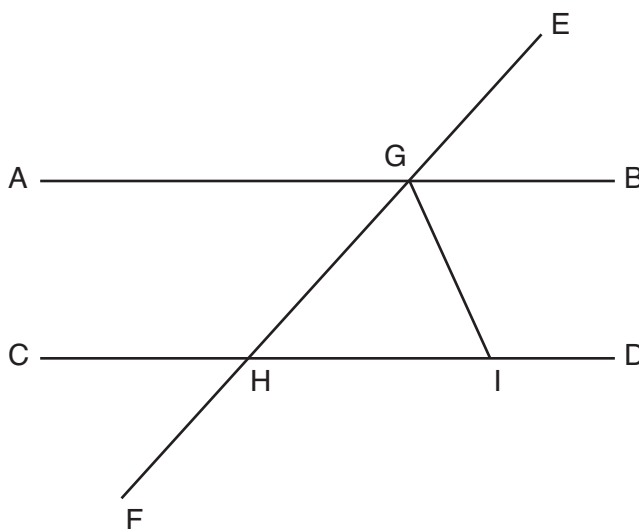
30 After a reflection over a line, $\triangle A'B'C'$ is the image of $\triangle ABC$. Explain why triangle ABC is congruent to triangle $A'B'C'$.

- 31** A flagpole casts a shadow 16.60 meters long. Tim stands at a distance of 12.45 meters from the base of the flagpole, such that the end of Tim's shadow meets the end of the flagpole's shadow. If Tim is 1.65 meters tall, determine and state the height of the flagpole to the *nearest tenth of a meter*.

Part III

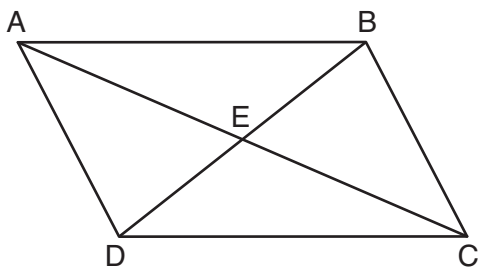
Answer all 3 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [12]

- 32 In the diagram below, $\overleftrightarrow{EF}$ intersects $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$ at G and H , respectively, and $\overleftrightarrow{GI}$ is drawn such that $\overline{GH} \cong \overline{IH}$.



If $m\angle EGB = 50^\circ$ and $m\angle DIG = 115^\circ$, explain why $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$.

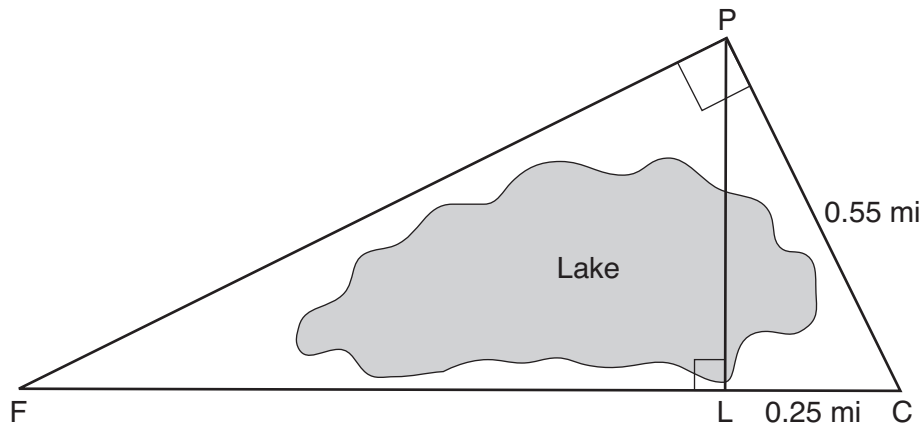
33 Given: Quadrilateral $ABCD$ is a parallelogram with diagonals $\overline{AC}$ and $\overline{BD}$ intersecting at E



Prove: $\triangle AED \cong \triangle CEB$

Describe a single rigid motion that maps $\triangle AED$ onto $\triangle CEB$.

- 34 In the diagram below, the line of sight from the park ranger station, P , to the lifeguard chair, L , on the beach of a lake is perpendicular to the path joining the campground, C , and the first aid station, F . The campground is 0.25 mile from the lifeguard chair. The straight paths from both the campground and first aid station to the park ranger station are perpendicular.



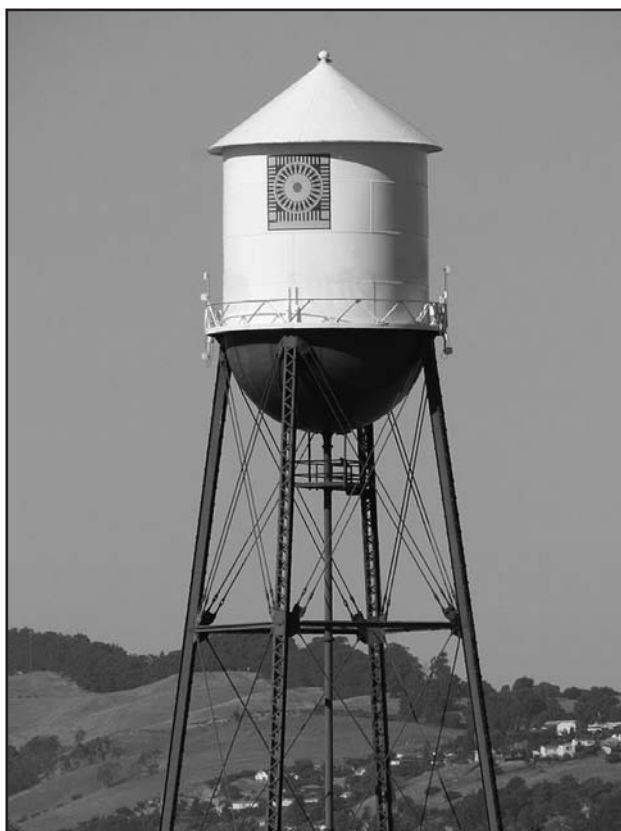
If the path from the park ranger station to the campground is 0.55 mile, determine and state, to the *nearest hundredth of a mile*, the distance between the park ranger station and the lifeguard chair.

Gerald believes the distance from the first aid station to the campground is at least 1.5 miles. Is Gerald correct? Justify your answer.

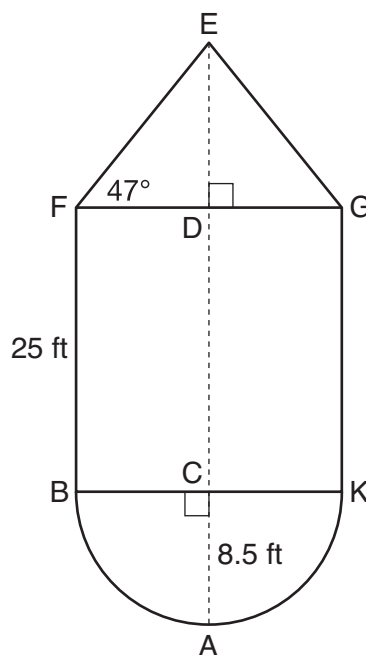
Part IV

Answer the 2 questions in this part. Each correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [12]

- 35 The water tower in the picture below is modeled by the two-dimensional figure beside it. The water tower is composed of a hemisphere, a cylinder, and a cone. Let C be the center of the hemisphere and let D be the center of the base of the cone.



Source: <http://en.wikipedia.org>



Question 35 is continued on the next page.

Question 35 continued

If $AC = 8.5$ feet, $BF = 25$ feet, and $m\angle EFD = 47^\circ$, determine and state, to the *nearest cubic foot*, the volume of the water tower.

The water tower was constructed to hold a maximum of 400,000 pounds of water. If water weighs 62.4 pounds per cubic foot, can the water tower be filled to 85% of its volume and *not* exceed the weight limit? Justify your answer.

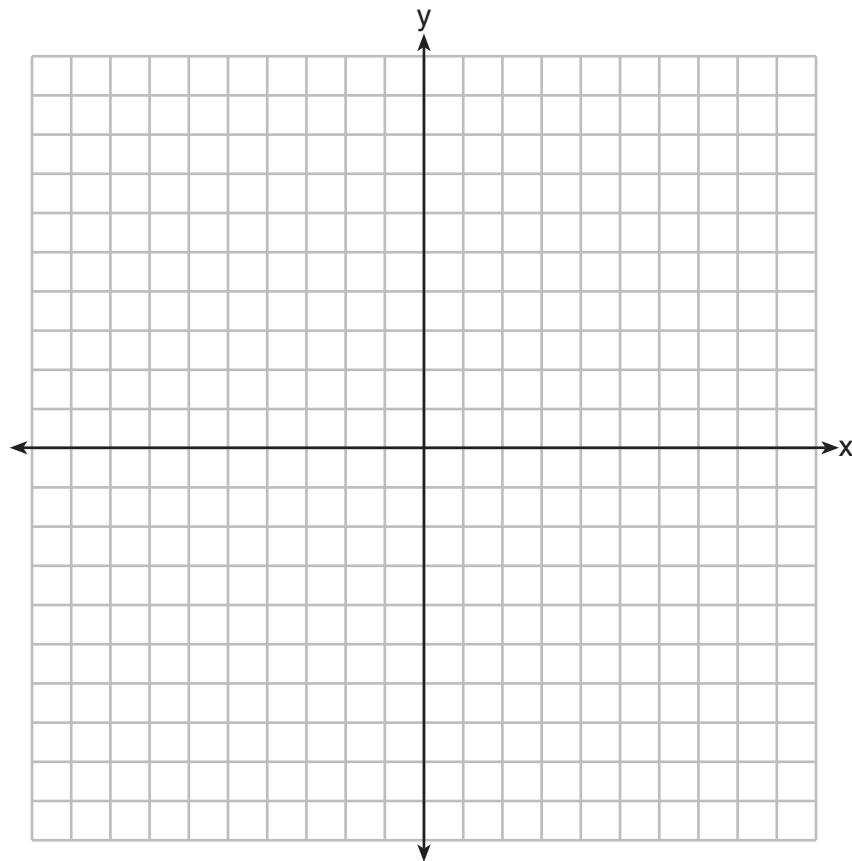
- 36** In the coordinate plane, the vertices of $\triangle RST$ are $R(6, -1)$, $S(1, -4)$, and $T(-5, 6)$.
Prove that $\triangle RST$ is a right triangle.
[The use of the set of axes on the next page is optional.]

State the coordinates of point P such that quadrilateral $RSTP$ is a rectangle.

Question 36 is continued on the next page.

Question 36 continued

Prove that your quadrilateral $RSTP$ is a rectangle.
[The use of the set of axes below is optional.]



If the student’s responses for the multiple-choice questions are being hand scored prior to being scanned, the scorer must be careful not to make any marks on the answer sheet except to record the scores in the designated score boxes. Marks elsewhere on the answer sheet will interfere with the accuracy of the scanning.

Part I

Allow a total of 48 credits, 2 credits for each of the following. Allow credit if the student has written the correct answer instead of the numeral 1, 2, 3, or 4.

(1) 4	(9) 1	(17) 1
(2) 4	(10) 1	(18) 1
(3) 3	(11) 3	(19) 2
(4) 4	(12) 4	(20) 1
(5) 3	(13) 4	(21) 4
(6) 2	(14) 2	(22) 1
(7) 3	(15) 3	(23) 2
(8) 1	(16) 2	(24) 3

Updated information regarding the rating of this examination may be posted on the New York State Education Department’s web site during the rating period. Check this web site at: <http://www.p12.nysed.gov/assessment/> and select the link “Scoring Information” for any recently posted information regarding this examination. This site should be checked before the rating process for this examination begins and several times throughout the Regents Examination period.

Beginning in June 2015, the Department is providing supplemental scoring guidance, the “Model Response Set,” for the Regents Examination in Geometry (Common Core). This guidance is intended to be part of the scorer training. Schools should use the Model Response Set along with the rubrics in the Scoring Key and Rating Guide to help guide scoring of student work. While not reflective of all scenarios, the Model Response Set illustrates how less common student responses to constructed-response questions may be scored. The Model Response Set will be available on the Department’s web site at: <http://www.nysedregents.org/Geometrycc/>.

**Map to the Common Core Learning Standards
Geometry (Common Core)
June 2015**

Question	Type	Credits	Cluster
1	Multiple Choice	2	G-GMD.B
2	Multiple Choice	2	G-CO.B
3	Multiple Choice	2	G-GPE.B
4	Multiple Choice	2	G-CO.A
5	Multiple Choice	2	G-SRT.C
6	Multiple Choice	2	G-GMD.B
7	Multiple Choice	2	G-MG.A
8	Multiple Choice	2	G-SRT.B
9	Multiple Choice	2	G-GPE.B
10	Multiple Choice	2	G-CO.A
11	Multiple Choice	2	G-SRT.B
12	Multiple Choice	2	G-SRT.C
13	Multiple Choice	2	G-CO.C
14	Multiple Choice	2	G-GPE.A
15	Multiple Choice	2	G-SRT.B
16	Multiple Choice	2	G-SRT.B
17	Multiple Choice	2	G-CO.C
18	Multiple Choice	2	G-SRT.A
19	Multiple Choice	2	G-MG.A
20	Multiple Choice	2	G-C.A
21	Multiple Choice	2	G-SRT.B
22	Multiple Choice	2	G-SRT.A
23	Multiple Choice	2	G-GMD.A
24	Multiple Choice	2	G-CO.B
25	Constructed Response	2	G-CO.D
26	Constructed Response	2	G-CO.C

27	Constructed Response	2	G-GPE.B
28	Constructed Response	2	G-SRT.C
29	Constructed Response	2	G-C.B
30	Constructed Response	2	G-CO.B
31	Constructed Response	2	G-SRT.B
32	Constructed Response	4	G-CO.C
33	Constructed Response	4	G-CO.C
34	Constructed Response	4	G-SRT.C
35	Constructed Response	6	G-MG.A
36	Constructed Response	6	G-GPE.B

Part II

For each question, use the specific criteria to award a maximum of 2 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(25) [2] A correct construction is drawn showing all appropriate arcs.

[1] Appropriate work is shown, but one construction error is made.

[0] A drawing that is not an appropriate construction is shown.

or

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(26) [2] A correct explanation is written.

[1] Appropriate work is shown, but one computational error is made.

or

[1] Appropriate work is shown, but an incomplete explanation is written.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(27) [2] $(-2, -3)$, and correct work is shown.

[1] Appropriate work is shown, but one computational or graphing error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] Appropriate work is shown to find -2 and -3 , but the answer is not written as coordinates.

or

[1] Appropriate work is shown, and point P is graphed correctly, but the coordinates are not stated or are stated incorrectly.

or

[1] $(-2, -3)$, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(28) [2] 23, and correct work is shown.

[1] Appropriate work is shown, but one computational or rounding error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] $\sin x = \frac{4.5}{11.75}$ or an equivalent trigonometric equation is written, but no further correct work is shown.

or

[1] 23, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(29) [2] 120 or $\frac{2\pi}{3}$, and correct work is shown.

[1] Appropriate work is shown, but one computational error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] 120 or $\frac{2\pi}{3}$, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(30) [2] A correct explanation is written.

[1] An incomplete explanation is written.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(31) [2] 6.6, and correct work is shown.

[1] Appropriate work is shown, but one computational or rounding error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] 6.6, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

Part III

For each question, use the specific criteria to award a maximum of 4 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

- (32) [4] A complete and correct explanation is written.
- [3] Appropriate work is shown, but one computational error is made. An appropriate explanation is written.
- or***
- [3] At least five correct angles are stated, but the explanation is incomplete.
- [2] Appropriate work is shown, but two or more computational errors are made. An appropriate explanation is written.
- or***
- [2] Appropriate work is shown, but one conceptual error is made.
- or***
- [2] At least five correct angles are stated, but the explanation is incorrect or is missing.
- [1] Appropriate work is shown, but one conceptual error and one computational error are made.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(33) [4] A complete and correct proof is written that includes a concluding statement, and a correct single rigid motion is stated.

[3] A proof is written that demonstrates a thorough understanding of the method of proof and contains no conceptual errors, but one statement and/or reason is missing or is incorrect. A correct single rigid motion is stated.

or

[3] A complete and correct proof that includes a concluding statement is written, but the rigid motion is not stated or is stated incorrectly.

[2] A proof is written that demonstrates a thorough understanding of the method of proof and contains no conceptual errors, but one statement and/or reason is missing or is incorrect. The rigid motion is stated incorrectly or is missing.

or

[2] A proof is written that demonstrates a good understanding of the method of proof and contains no conceptual errors, but two statements and/or reasons are missing or are incorrect. A correct rigid motion is stated.

or

[2] A proof is written that demonstrates a good understanding of the method of proof, but one conceptual error is made.

[1] A correct rigid motion is stated, but no further correct work is shown.

or

[1] Some correct relevant statements about the proof are made, but three or four statements and/or reasons are missing or are incorrect.

[0] The “given” and/or “prove” statements are written, but no further correct relevant statements are written.

or

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (34) [4] 0.49, and correct work is shown. No, with a correct justification, is written.
- [3] Appropriate work is shown, but one computational or rounding error is made.
- [2] Appropriate work is shown, but two or more computational or rounding errors are made.
- or***
- [2] Appropriate work is shown, but one conceptual error is made.
- or***
- [2] Correct work is shown to find 0.49 or No, with a correct justification. No further correct work is shown.
- [1] Appropriate work is shown, but one conceptual error and one computational error are made.
- or***
- [1] A correct equation is written to find the distance between the park ranger station and the lifeguard chair, or a correct equation is written to find the distance between the first aid station and the campground, but no further correct work is shown.
- or***
- [1] 0.49, but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
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Part IV

For each question, use the specific criteria to award a maximum of 6 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(35) [6] 7650 and No, and correct work is shown.

[5] Appropriate work is shown, but one computational or rounding error is made.

or

[5] Correct work is shown to find 7650 and 85% of the water tower's volume, but no further correct work is shown.

[4] Appropriate work is shown, but two computational or rounding errors are made.

or

[4] Appropriate work is shown, but one conceptual error is made.

or

[4] Correct work is shown to find the total volume of the tower, 7650, but no further correct work is shown.

[3] Appropriate work is shown, but three or more computational or rounding errors are made.

or

[3] Appropriate work is shown, but one conceptual error and one computational or rounding error are made.

or

[3] Correct work is shown to find the volume of each geometric component of the water tower, but no further correct work is shown.

[2] Correct work is shown to find the volume of the cone, but no further correct work is shown.

or

[2] Appropriate work is shown, but one conceptual error and two or more computational or rounding errors are made.

or

[2] Appropriate work is shown, but two conceptual errors are made.

[1] Appropriate work is shown, but two conceptual errors and one computational or rounding error are made.

or

[1] Correct work is shown to find the volume of the hemisphere and the volume of the cylinder or the height of the cone, but no further correct work is shown.

or

[1] 7650 and No, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (36) [6] Correct work is shown to prove $\triangle RST$ is a right triangle. $P(0,9)$ is stated, and correct work is shown to prove $RSTP$ is a rectangle.
- [5] Appropriate work is shown, but one computational or graphing error is made.
- or**
- [5] Appropriate work is shown, but one concluding statement is missing or is incorrect.
- or**
- [5] Correct proofs are written, but $P(0,9)$ is not stated.
- [4] Appropriate work is shown, but two computational or graphing errors are made.
- or**
- [4] Appropriate work is shown, but one conceptual error is made.
- or**
- [4] Correct work is shown to find $P(0,9)$ and prove quadrilateral $RSTP$ is a rectangle.
- [3] Appropriate work is shown, but three or more computational or graphing errors are made.
- or**
- [3] Appropriate work is shown, but one conceptual error and one computational or graphing error are made.
- or**
- [3] Correct work is shown to prove $\triangle RST$ is a right triangle, and $P(0,9)$ is stated. No further correct work is shown.
- [2] Appropriate work is shown, but one conceptual error and two or more computational or graphing errors are made.
- or**
- [2] Appropriate work is shown, but two conceptual errors are made.
- or**
- [2] Correct work is shown to prove $\triangle RST$ is a right triangle, but no further correct work is shown.

[1] Appropriate work is shown, but two conceptual errors and one computational or graphing error are made.

or

[1] $P(0,9)$ is stated, but no further correct work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
